

Enterprise E-Commerce REST API Testing Strategy

Project Overview

This project validates an enterprise-style e-commerce REST API using Postman, Newman, JavaScript assertions, and GitHub Actions CI/CD.

Objectives

Verify functional correctness, business rules, data integrity, performance, and regression stability.

Scope

Authentication, Products, Users, Carts, CRUD operations, Negative Testing, Data Integrity Validation, Performance Validation.

Test Approach

Risk-based API testing using reusable Postman collections, collection variables, QA environment, automated assertions, Newman CLI, and GitHub Actions.

Test Types

Functional, Regression, Smoke, Negative, Data Integrity, Response Validation, Performance Validation.

Environment

QA environment with GitHub Actions Secrets for credentials and Collection Variables for configuration.

Entry Criteria

API available, collection configured, QA environment prepared.

Exit Criteria

All critical tests pass, HTML/JUnit reports generated, no Sev-1/Sev-2 defects open.

Risks & Mitigations

API downtime, unstable data, credential issues; mitigated with retries, secrets, and isolated test data.

Deliverables

Postman Collection, QA Environment, Newman HTML/JUnit reports, GitHub Actions pipeline, documentation.

Tools

Postman v12, Newman, JavaScript, GitHub Actions, Git, REST, JSON.